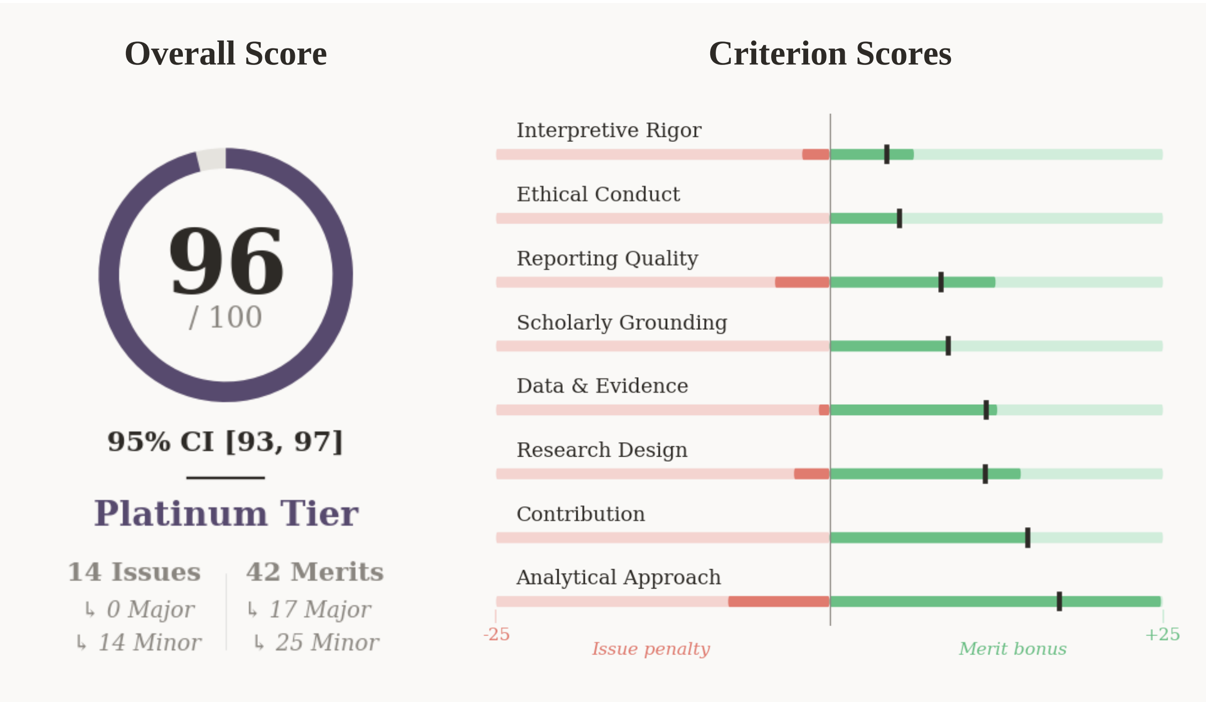
explore science

Replay Is a Family of Assertions: Fork-Safety in Event-Sourced Agent Runtimes

Review Report

August 28, 2026
explorescience.ai

Share with co-authors



Explore Science is all about making your paper be the best it can possibly be. Our goal is to improve science, your science, and help all of us in finding academic truth. The feedback provided in this review is purely for educational purposes only and we accept no liability or responsibility for any suggestions made. We recommend evaluating each point of feedback critically before integrating it into your paper.AI is not perfect, and can make mistakes.

Contents

1. Manuscript Summary

Researchers at activegraph.ai demonstrate that claims of "deterministic replay" and "cheap forking" in event-sourced agent runtimes conflate distinct operational guarantees. While copying an event log is trivial, irreversible external interactions and schedule-sensitive executions compromise semantic replication. By formalizing property-relative fork assessments across 5,540 logs, the authors reveal how split execution boundaries invalidate environmental continuation. This reframes agent branching from a storage optimization into a strict semantic contract, obligating systems to explicitly track effect footprints to guarantee safe state recovery.

2. Scoring

This manuscript was scored across eight criteria, from research design to interpretive rigour. Both issues and merits contribute to the score. Each point is classified as major or minor based on its nature and potential impact on the research.

> *» Learn how scoring works at explorescience.ai/scoring*

Relative Performance Metrics - Coming Soon

Field-specific percentile distributions are not yet available for your discipline. We can let you know when they are - register your interest.

3. Summary of Notable Merits

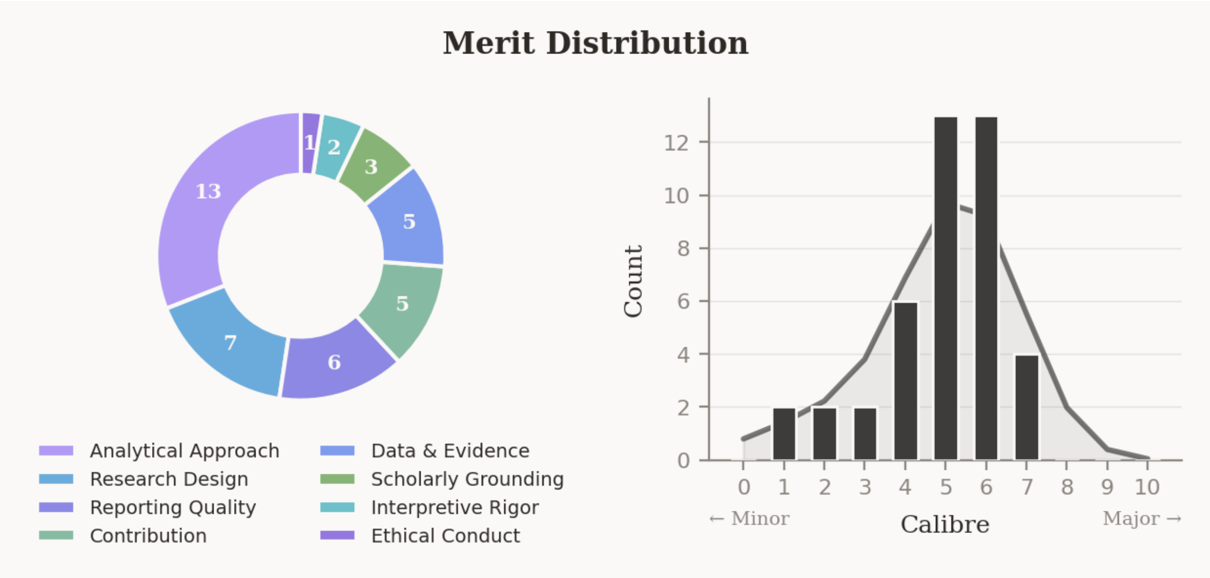
The review identifies this manuscript as a work of notably high analytical and conceptual calibre, whose strengths concentrate around two dominant patterns: the construction of formally precise, multi-dimensional definitions that replace field-standard overloaded terms, and an empirical validation discipline that holds every quantitative claim to exact, independently verifiable accounting rather than probabilistic approximation. Together, these patterns exceed contemporary norms in event-sourced agent runtime research, where binary pass/fail replay verification and conflated single-tier guarantees remain standard, and they materially strengthen the inferential basis for the paper's central reframing of fork safety as a semantic contract.

The three most notable merits identified were:

- A three-dimensional effect taxonomy (external footprint × replay source × lifecycle status) that enables precise cut-level safety assessment, directly surpassing the flat Pure/Idempotent/Cached/OneShot partitions prevalent in deployed durable execution platforms.
- Voluntary self-quarantine of an unverifiable empirical sub-study from all headline claims, where prevailing practice routinely foregrounds findings regardless of independent verifiability.
- A dual-track validation design pairing observed forks with exhaustive all-cut enumeration, preventing positive evidence from masking latent hazards that practice happened to avoid, unlike the binary replay verification standard in comparable work.

Collectively, these strengths establish the manuscript as a rigorous and credible contribution to the formal foundations of agent execution continuity.

» *View all merits online*



4. Review Synopsis

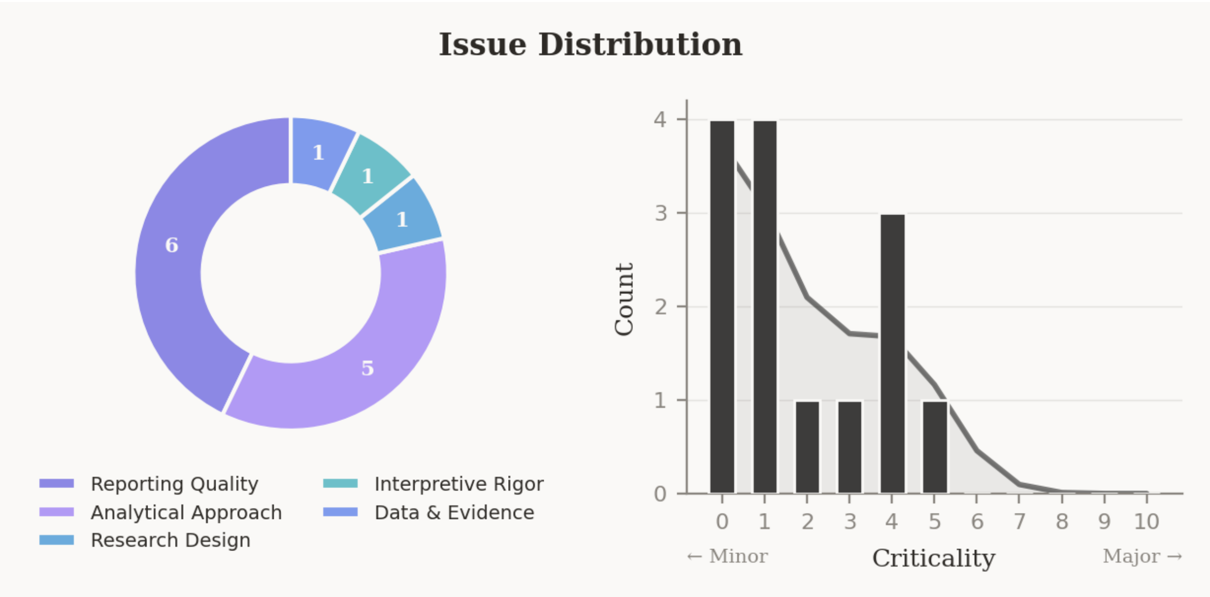
The manuscript is in strong overall condition, with revision attention concentrated in a coherent cluster of analytical precision concerns rather than any structural deficiency. The dominant theme across the review is internal consistency within the formal definitions: several permutations and coupling decisions in the core definitional apparatus produce verdicts or constraints that sit in tension with the paper's own stated goals, exemplified by the fail-closed rule for a specific Unknown/Recorded/Committed permutation and the scope of blocking conditions in Definition 6. A secondary, lighter theme concerns reporting completeness in the methods sections, where several terms central to verifying quantitative claims appear without explicit definition. Together, these patterns reduce confidence in the precision of the formal guarantees as currently stated - a concern that bears directly on the paper's central reframing of fork safety as a semantic contract, which depends on those definitions being mutually consistent.

The three highest-priority issues identified in the review were:

- the over-restrictive Unsound verdict for the Unknown/Recorded/Committed permutation in the ExternalContinuation matrix **(#B1)**
- the globally applied blocking rule for discarded OneShot effects in Definition 6 **(#B2)**
- the hard coupling between ExternalContinuation and StrictExecutionReplay creating a dimensional orthogonality inconsistency **(#B3)**

Resolving the internal consistency of Definitions 4, 5, and 6 in tandem would most substantively strengthen the manuscript's formal contribution.

> *» The 10 most critical issues are detailed below. View all online.*



5. Comments

The 10 highest-priority issues are detailed below. View the complete list online →

5.1 Methods

#B1. Fail-Closed Treatment of Unknown Footprint in ExternalContinuation Assessment

Critique: ♦ ANALYTICAL APPROACH

The ExternalContinuation verdict matrix warrants attention for the Unknown + Recorded + Committed permutation, where the current fail-closed rule produces an Unsound verdict that appears over-restrictive relative to Definition 5's stated purpose.

Justification & Consequence: − MINOR

Definition 5 explicitly does not assert world-state equivalence - only that the child can continue safely. For an effect with a Committed lifecycle and a Recorded replay source, the runtime possesses everything required to serve the result without any live oracle call, satisfying the zero-re-execution goal. The sole uncertainty is the ontological footprint of the original execution, which concerns past world-state, not future child safety. Conflating footprint uncertainty with re-execution unsafety collapses two distinct concerns, and the paper provides no explicit argument for why this conflation is necessary under Definition 5, leaving the over-conservatism unjustified.

Relevant Section:

Definition 5 (ExternalContinuation) and Table 3 fork conditions (pp. 8-9); Section 4.1 orthogonal effect dimensions and Table 2 (p. 8); Section 6.3 fail-closed rule description (p. 13).

Framework for Addressing:

- In Table 3 and the surrounding discussion, add a row or note explicitly addressing the Unknown + Recorded + Committed case, arguing either that the Unsound verdict is intentional (with justification) or revising it to Conditional - permitting the child to continue while flagging that world-state claims cannot be made.
- If revised to Conditional, update Section 4.3's prose to clarify that Unknown footprint with terminal Committed lifecycle distinguishes future execution safety from past ontological uncertainty under Definition 5.

#B2. Scoping of CounterfactualWorld Blocking Rule to Declared Observable Set Ω

Critique: ♦ ANALYTICAL APPROACH

Definition 6's blocking rule for CounterfactualWorld(Ω) requires revision to condition the block on whether a discarded OneShot effect's observable falls within the declared scope Ω, rather than applying globally.

Justification & Consequence: – MINOR

Ω is explicitly parameterised as a declared set of external observables, yet the stated rule - "discarded committed OneShot or unknown effects block the claim" - is unscoped. An effect whose observable lies entirely outside Ω (e.g., a telemetry ping when Ω covers only financial ledger entries) should not block a CounterfactualWorld(Ω) claim about that ledger. The Section 6.3 adapter already applies profile-scoped logic for Sound continuation verdicts, but this scoping is not carried through to Definition 6, producing conservatively inflated Unsound and Conditional counts in Table 6 and reducing the framework's practical utility for systems with heterogeneous, targeted observability requirements.

Relevant Section:

Definition 6 (CounterfactualWorld) (p. 9); Table 3 fork conditions (p. 9); Section 6.3 profile declaration discussion (p. 13); Table 6 bridge cut verdicts (p. 16).

Framework for Addressing:

- Revise Definition 6 to explicitly scope the blocking condition: "Discarded committed OneShot or unknown effects whose observables intersect Ω block the claim; effects with observables entirely outside Ω do not."
- Update Table 3 and Section 9.2's fork API discussion to reflect that CounterfactualWorld(Ω) verdicts depend jointly on the effect's footprint and its membership in Ω.
- Note in Section 7.2 how this revised scoping would affect the bridge study's Conditional/Unsound counts.

#B3. Coupling of ExternalContinuation to Zero-Call StrictExecutionReplay for Idempotent Effects

Critique: ♦ ANALYTICAL APPROACH

The hard coupling of ExternalContinuation (Definition 5) to the zero-live-call requirement of StrictExecutionReplay (Definition 4) creates an internal inconsistency with the paper's claim that external footprint and replay source are orthogonal dimensions.

Justification & Consequence: – MINOR

Definition 5 requires strict replay as a prerequisite, mandating zero oracle or tool calls regardless of footprint. For an Idempotent + Uncaptured permutation, this forces refusal of a live re-execution that is, by the framework's own definition of Idempotent, safe to repeat under an explicit key - a property that distributed systems exploit precisely to tolerate re-execution safely (Bernstein & Newcomer, 2009). Section 4.1 asserts the three dimensions are orthogonal, yet this coupling collapses footprint and replay-source into a single decision axis for this combination, producing a verdict that is more restrictive than the declared semantics of Idempotent footprint require.

Relevant Section:

Definition 4 (StrictExecutionReplay) and Definition 5 (ExternalContinuation) (pp. 8-9); Section 4.1 orthogonal effect dimensions and Table 2 (p. 8); Table 3 fork conditions (p. 9).

Framework for Addressing:

- In Definition 5 or Table 3, add an explicit clause: for an Idempotent + Uncaptured + Committed permutation, ExternalContinuation may be assessed as Conditional (live re-execution permitted under the idempotency key) rather than Unsound, with the obligation documented in the fork receipt.
- Add a sentence in Section 4.1 or 4.2 acknowledging that the Idempotent footprint dimension partially relaxes the zero-call constraint, clarifying the precise boundary at which the coupling applies.

#B4. Non-Atomic External API Failures and Request/Outcome Cut Coverage

Critique: • RESEARCH DESIGN

The Request/Outcome cut taxonomy (Figure 4) and Definition 6's blocking conditions require a discussion of non-atomic external API failures, which can produce partial external mutations under a Failed lifecycle and fall outside the current blocking logic.

Justification & Consequence: - MINOR

Definition 6 restricts CounterfactualWorld blocks to "committed OneShot or unknown effects." In distributed systems, non-atomic operations - batch writes, multi-step remote processes - can mutate external state before returning a failure response, meaning the interpreter logs a Failed lifecycle while the world has partially changed (Gray & Reuter, 1992). This case falls between the k_1 and k_2 boundaries of Figure 4 yet is assigned a terminal Failed outcome, which the current blocking rule treats as safe. The resulting gap could produce a falsely Sound CounterfactualWorld verdict for a fork whose discarded suffix contains an externally partially-committed Failed effect.

Relevant Section:

> *Definition 6 (CounterfactualWorld) and Table 3 fork conditions (pp. 9); Figure 4 request/outcome cut regions (p. 10); Section 4.1 effect dimensions including OneShot/Compensatable description (p. 8).*

Framework for Addressing:

- Add a brief paragraph in Section 4.4 (or Section 10 Limitations) acknowledging that the $k_0/k_1/k_2$ taxonomy assumes atomic external API responses; non-atomic partial-commit scenarios where a Failed outcome co-occurs with external state mutation are outside the current model.
- Clarify that the framework's conservatism in treating Unknown footprint (rather than just Failed) provides partial mitigation, and note this as a boundary condition requiring explicit recovery policy.

#B5. Temporal Side-Effects of Compensation in CounterfactualWorld Assessment

Critique: • INTERPRETIVE RIGOR

The treatment of Compensatable effects in Definition 6 could be strengthened by acknowledging that successful compensation does not guarantee temporal-window counterfactual equivalence.

Justification & Consequence: MINOR

Conditioning CounterfactualWorld on "reconciliation or successful compensation" assumes compensation perfectly restores the environment, but saga literature acknowledges that compensation does not retroactively erase the interval during which the original state was observable (Garcia-Molina & Salem, 1987). During that window, secondary side-effects in third-party systems (quota triggers, dependent webhooks, cron jobs) may be uncompensatable. The framework can issue a Conditional verdict for a Compensatable effect, yet the counterfactual claim may remain unsound due to temporal exposure the compensation cannot retract - a limitation the authors note in passing for sagas but do not apply to the Compensatable dimension's world-level guarantees.

Relevant Section:

Definition 6 (CounterfactualWorld) (p. 9); Table 3 fork conditions (p. 9); Section 4.1 Compensatable dimension in Table 2 (p. 8); Section 8 saga discussion (p. 17).

Framework for Addressing:

- In Definition 6 or Section 4.1, add a qualifying sentence noting that Conditional verdicts for Compensatable effects assume compensation is both successful and timely; temporal windows during which compensated state was observable may introduce secondary effects outside the model's scope.
- Optionally add this as a bullet in Section 10 Limitations alongside the saga reference already present in Section 8.

#B6. Consistency Between FIRE-ACTIVATION Rule and Same-Trigger State-Observation Claim

Critique: REPORTING QUALITY

The FIRE-ACTIVATION transition rule in Figure 2 raises a consistency concern with the prose claim in Section 2.3 that later activations observe state changes from earlier activations on the same triggering event.

Justification & Consequence: MINOR

The FIRE-ACTIVATION rule holds state s invariant - events materialized by one activation enter Q' as pending and are only projected into state upon a subsequent PROCESS-EVENT step. Multiple activations sharing a triggering event therefore read identical s, not an updated state. The paper should clarify that the described observation ordering operates across PROCESS-EVENT steps rather than within a single activation batch for the same trigger, since the key non-confluence witness (Counterexample 3) in fact relies on distinct triggering events and is unaffected by this narrow inconsistency.

Relevant Section:

Section 2.3 scheduling description including the sequential firing claim (p. 4); Figure 2 FIRE-ACTIVATION and PROCESS-EVENT rules (p. 5).

Framework for Addressing:

- Revise the Section 2.3 prose to read: "A behavior activated by a later-processed event observes state changes committed by earlier PROCESS-EVENT steps, including those materialised from prior activations" - clarifying that the ordering seam operates across event-processing steps, not within a single activation batch for the same trigger.

- Optionally add a brief parenthetical in Figure 2's caption noting that intra-trigger state visibility requires an intervening PROCESS-EVENT step.

#B7. Definition of 'Null Divergence' as a Boundary Grade Prerequisite

Critique: REPORTING QUALITY

A minor clarification would help readers verify the 24 Boundary verdicts in Table 6: 'null divergence,' cited in Section 5.1 as a prerequisite for the bridge Boundary derivation, is not defined anywhere in the paper.

Justification & Consequence: MINOR

The term appears in no other section - not in the event vocabulary (Section 2.2), effect descriptor (Table 2), or conformance vectors - yet it is a stated prerequisite for a grade that underpins the bridge study's demonstration that Boundary evidence cannot promise world-state equivalence. The bridge study is scoped as illustrative and excluded from headline claims, so the practical impact is limited, but the undefined prerequisite reduces the transparency the paper claims for its grade computation.

Relevant Section:

Section 5.1 grade computation prerequisites (p. 11); Table 6 bridge Boundary verdicts (p. 16).

Framework for Addressing:

- Add a parenthetical definition in Section 5.1, e.g.: "an explicit null divergence (a log-level assertion that no unresolved behavioral divergence was detected during reconstruction)."
- Alternatively, include null divergence as a row in Table 2 or as a named event kind in Section 2.2 if it belongs to the closed vocabulary.

#B8. Formal Definition of 'Hazard Event' in the Grade Vocabulary

Critique: REPORTING QUALITY

A minor clarification would help readers verify Counterexample 4: 'hazard event' is used to establish grade non-monotonicity but is not defined within the paper's closed event or effect vocabulary.

Justification & Consequence: MINOR

The term appears in Section 5.2 and in the named regression fixtures (Section 6.2) without entry in Section 2.2's closed event sum, Table 2's effect dimensions, or the lifecycle values. Non-monotonicity of grades is an operationally important property - foregrounded in Table 1 and Section 5.3 - and the counterexample is its only formal witness. The contextual meaning (an event that introduces a grade-lowering blocker) is clear, but the absence of a formal definition in the executable vocabulary slightly undermines the precision the paper claims for its contract-based approach.

Relevant Section:

Section 5.2 Counterexample 4 (p. 11); Section 6.2 named regression fixtures (p. 12-13); Section 2.2 event kinds (p. 3).

Framework for Addressing:

- Add 'HazardEvent' (or equivalent) to the closed event kind sum in Section 2.2 with a one-sentence definition specifying that it introduces a grade-lowering blocker when appended to a log.
- Alternatively, add a parenthetical in Counterexample 4: e.g., "a hazard event (a log entry whose recognized type is mapped to a grade blocker in the adapter)."

5.2 Results and Analysis

#C1. All-Cut Enumeration on Intra-Source-Event Normalization Positions in the Bridge Study

Critique: ♦ ANALYTICAL APPROACH

The all-cut enumeration over the bridge study's normalized event stream generates synthetic intra-expansion cut positions - between derived facts from the same atomic source event - that do not correspond to physically executable branch points in the source log.

Justification & Consequence: − MINOR

The authors commendably document the 1-to-N expansion (2,592 2,640 events) and the resulting cut count (2,664 in Table 6). At an intra-expansion cut between a fresh-reconstruction fact and its paired mediated-boundary fact, the prefix satisfies neither the full Boundary prerequisite nor valid ExternalContinuation conditions, yet these positions are included in the reported totals. The non-contiguity blocker addresses identity integrity but does not reclassify these positions as non-executable artifacts. The reported cut counts therefore include positions that are artefacts of normalisation rather than genuine fork candidates, a distinction the paper does not draw.

Relevant Section:

Section 6.3 normalization and multi-fact expansion (p. 13); Section 7.2.2 bridge cut verdicts and Table 6 (p. 16).

Framework for Addressing:

- Add a sentence in Section 7.2.2 noting that cuts between co-derived facts from the same source event are normalization artifacts and do not represent physically executable fork points, clarifying the denominator of the 2,664 reported cuts.
- Optionally report source-event-boundary cut counts separately from intra-expansion positions to distinguish the two populations.

#C2. Empirical Coverage of the Environment Attestation Discharge Pathway

Critique: ✚ DATA & EVIDENCE

The 160,076 Sound ExternalContinuation verdicts in Table 5 validate only the trivial no-inherited-effect case, leaving the attestation-discharge mechanism - the operationally significant pathway for post-LLM-call forks - without empirical support in the public corpus.

Justification & Consequence: MINOR

Definition 5 specifies that the environment premise can be discharged by a snapshot identifier or fresh-environment receipt; absent such attestation, the verdict is Conditional. Inspection of the corpus shows that the 160,076 Sound cuts precede the first classified LLM request and therefore never require attestation. The paper acknowledges in Section 10 that no observed fork crosses an oracle boundary, but does not connect this explicitly to what the Sound ExternalContinuation counts in Table 5 do and do not validate. Synthetic fixtures (Section 6.2) exercise the rule mechanically, but real-corpus validation of the attestation pathway remains absent - a limit the authors note.

Relevant Section:

> *Definition 5 (ExternalContinuation) (p. 8); Table 5 all-cut verdicts (p. 15); Section 7.1.3 observed forks discussion (p. 15); Section 10 fork coverage limitation (p. 21).*

Framework for Addressing:

- In Section 7.1.2 or 7.3, add a sentence clarifying that the 160,076 Sound ExternalContinuation cuts are pre-effect cuts where no attestation is required, and that empirical exercise of the attestation-discharge pathway awaits a corpus containing post-effect forks with explicit environment receipts.

6. References

Bernstein, P., & Newcomer, E. (2009). Principles of Transaction Processing. Morgan Kaufmann. https://doi.org/10.1016/b978-1-55860-623-4.x0001-7

Garcia-Molina, H., & Salem, K. (1987). Sagas. Proceedings of the 1987 ACM SIGMOD International Conference on Management of Data. https://doi.org/10.1145/38713.38742

Gray, J., & Reuter, A. (1992). Transaction Processing: Concepts and Techniques. Morgan Kaufmann.